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Section : A

Subject : Computer Networks

Program: BSAI

Course Code: CS-307

Assignment Title: Transmission Media and Physical Topologies

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Part # 1 :

Transmission Media :

1. Computers inside a computer lab are close together and the network must be low cost and easy to install.

Suitable Media : Unshielded Twisted Pair (UTP) cable.

Reason : UTP cables are the most affordable and flexible option for short-distance, indoor networking. They are very easy to install and plug into standard computers, making them the perfect budget-friendly choice for a single lab environment.

2. The university wants to connect two campuses 3 km apart with very high speed and minimum signal loss.

Suitable Media: Optical fiber

Reason : Optical fiber uses light to transmit data, which allows it to carry huge amounts of information at extremely high speeds. It is completely immune to electrical interference and can easily travel distance like 3 km without losing signal quality.

3.

The campus wifi must provide wireless internet access to students in open areas like gardens and grounds.

Suitable Media : Radio Waves

Reason : Radio waves are omnidirectional, meaning they travel in all directions from the source (like a Wi-Fi router). They can easily cover wide, open outdoor spaces and penetrate through minor obstacles, making them ideal for standard Wi-Fi setups.

4.

A television studio needs very

clear video signals without electrical noise interference.

Suitable Media: Coaxial Cable

Reason: Coaxial cables are built with a special metallic shield wrapped around the inner conductor. This shielding blocks out outside electrical noise and electromagnetic interference, ensuring that high-quality video signals remain clear and stable.

5. Two office buildings across a road must communicate wirelessly with high bandwidth point-to-point communication.

Suitable Media: Microwaves

Reason: Microwaves provide excellent high-bandwidth wireless communication, but they require a strict "line-of-sight" setup. Placing microwave antennas on the roofs of two buildings directly facing each other across a road perfectly meets this placement.

6. A t.v remote control sends signals to a television in the same room.

Suitable Media: Infrared.

Reason: Infrared relies on short-range, line-of-sight communication. It cannot pass through walls, which prevents remote controls from accidentally changing the channels on a TV in the next room over.

Part # 2:

Physical Topologies

1. A school computer lab where all computers connect to one central switch for easy management and troubleshooting.

Suitable topology: Star topology

Reason: In a star topology, every single device connects directly to a

central hub or switch. This makes it incredibly easy for administrators to manage the network and quickly figure out which specific cable or computer has an issue if something goes wrong.

2. A small temporary office where setup cost must be very low only a few computers need to be connected.

Suitable Topology : Bus topology

Reason: A bus topology connects all devices along a single main cable (the backbone). It uses the least amount of cabling and requires no central switch, making it the cheapest and easiest layout to set up for a short term need.

3. A bank network where communication must continue even if one connection fails.

Suitable Topology : Mesh Topology.

Reason: Financial institutions require strict reliability. In a Mesh topology, multiple devices are interconnected. If one specific cable or path breaks down, the data simply takes a different route to its destination, ensuring that bank stays online.

4.

A token based network system where each device should get equal chance to send data in order

Suitable Topology: Ring Topology

Reason: A ring topology connects computers in a closed loop. A "token" travels around this circle, and a computer can only send data when it holds the token. This guarantees that every device gets a fair, predictable turn to communicate without data collisions.

5.

A military communication system where every device must have backup paths to ensure high reliability.

Suitable topology: Mesh Topology

Reason: Military operations can't afford any network rundown. A full mesh topology connects every single device to every other device, providing maximum redundancy. Even if multiple links are destroyed, the network will still function.

6. A small office where if one computer fails, the rest of the network should continue working without interruption.

Suitable topology: Star topology

Reason: Because each computer in a star topology has its own dedicated cable connected to the central switch, one computer crashing or disconnecting will not affect any of the others. The rest will work normally.